

Load-dependent sacroiliac joint motion in 278 pelvic finite-element models

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Abstract

The relation between pelvic anatomy and sacroiliac joint (SIJ) response depends on both loading and the reference configuration used to measure motion. We reanalyzed saved finite-element fields from 278 pelvises (150 female, 128 male; age 16–91 years), comparing bilateral standing, unilateral standing and three localized force-pair proxies. Kinematics were measured between unloaded and loaded versions of the same mapped anatomy, separating anatomical variation from elastic displacement. Median bilateral translation was 1.058 mm under bilateral standing and 1.627 mm under unilateral standing; translation-magnitude asymmetry was 0.065 and 1.602 mm, respectively. The three force-pair cases yielded median translations of 0.154, 0.255, 0.574 mm. Age-, size- and morphology-adjusted female-minus-male translation coefficients ranged from -0.031 to +0.013 mm; 0 of the three translation comparisons survived FDR correction across six translation and rotation tests. Shape-only models retained 96.43–120.61% of full-model variance across the five reported endpoints, while material-only/full ratios reached 5.789%. These ratios are non-additive and concern bone density under fixed soft-tissue assumptions. Component contrasts describe absolute motion redistribution, not signed joint opening or birth-canal expansion. The resulting dataset supports comparisons among specified passive mechanical configurations; physiological, obstetric and clinical prediction requires independent validation.

Keywords: sacroiliac joint; pelvic functional anatomy; finite-element analysis; reference configuration; sex-related associations; pelvic architecture; variance ratios

1 Introduction

The sacroiliac joint (SIJ) transfers load between the trunk and lower limbs while allowing small relative movements of the sacrum and ilia. Articular geometry and ligamentous restraint contribute to this function; muscular compression also contributes in vivo (Vleeming et al., 2012; Snijders et al., 1993; Hammer et al., 2013). The relationship between pelvic stability and obstetric accommodation motivates questions about pelvic mechanics, but a static skeletal model cannot by itself establish an evolutionary trade-off or predict childbirth outcomes (Pavličev et al., 2020; Chen and Grimm, 2021).

Population-scale pelvic finite-element (FE) modeling is already established. In particular, Henyš and Hammer (2025) analyzed 281 pelvic reconstructions and related SIJ morphology to its mechanical environment. The present study therefore does not claim novelty for cohort size or for the general proposition that geometry matters. Its intended contribution is a within-subject comparison of five standardized load configurations, with separate absolute motion components and matched geometry/material variants. This design asks whether changing load location and direction changes the distribution of joint motion across anatomical axes, and whether these responses covary with sex and pelvic dimensions.

A second question concerns the extent to which a reference bone-density field can reproduce subject-specific kinematic responses. This is a restricted modeling question: holding cartilage and ligament properties fixed does not test their population variability. Recent work identifies ligament pretension as an influential SIJ modeling parameter (Heyland et al., 2025). Thus a comparison between geometry and bone-density variants cannot establish that geometry dominates all tissue properties.

We analyzed 278 imaging-derived pelvic models (150 female and 128 male; age 16–91 years) from the standardized BoneDat workflow (Henyš and Kuchař, 2025). Two standing configurations and three localized, parturition-motivated force pairs were compared at identical prescribed forces across subjects. The latter are mechanical probes rather than successive stages of labor. Joint motion was measured relative to each subject’s unloaded mapped anatomy, separating elastic displacement from inter-individual shape differences.

Hypotheses and scope

The following hypotheses organize a retrospective analysis; they were not preregistered and should not be interpreted as independent prospective confirmation.

1. **H1 (standing asymmetry):** Unilateral loading increases bilateral translation-magnitude asymmetry and overall joint motion relative to bilateral loading.
2. **H2 (component redistribution):** Relative to bilateral standing, LAB1 and LAB2 reduce the absolute CC translation component and increase the absolute ML component. LAB3 is assessed separately rather than assumed to share the same response. Absolute components do not establish signed joint opening or nutation direction.
3. **H3 (conditional sex association):** Female models have greater translation under the three LAB configurations after age adjustment; adjustment for size and the morphometric triad reduces this difference. These are conditional associations, not a causal mediation model.
4. **H4 (restricted material simplification):** Shape-only responses retain most of the full-model between-subject variance and agree with individual full-model predictions more closely than the material-only variant suggests. Variance ratios are not additive causal contributions.

Volume–motion associations and sex-by-volume interactions were exploratory secondary analyses. Their exponents describe conditional associations under fixed applied force, not universal physiological scaling laws.

2 Materials and methods

2.1 Cohort and provenance

The precomputed cohort comprised 278 pelvic models (150 female, 128 male), aged 16–91 years (mean $\pm$ SD: 53.9 ± 19.3 years), using the BoneDat morphology workflow (Henyš and Kuchař, 2025). Demographic, morphometric and size tables were joined by subject identifier. Eight morphometric variables and the kinematic outcomes were complete for the included cohort (Table 1). The previously recorded ethical approval is University Hospital Hradec Králové, reference 202411IO3P, with waiver of informed consent. Documentation of the cohort selection

and its overlap with the earlier 281-model study (Henyš and Hammer, 2025) is required to distinguish reuse of imaging from independent validation.

Table 1: Demographic and morphometric characteristics. Values are mean $\pm$ SD; dimensions are measured from mapped landmarks.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 $\pm$ 19.33	53.36 $\pm$ 19.74	54.52 $\pm$ 18.91	0
Inlet AP diameter (mm)	117.04 $\pm$ 10.52	121.04 $\pm$ 10.02	112.34 $\pm$ 9.07	0
Biacetabular width (mm)	125.81 $\pm$ 8.53	129.08 $\pm$ 8.21	121.98 $\pm$ 7.22	0
Biischiadic width (mm)	104.27 $\pm$ 9.33	109.87 $\pm$ 7.13	97.71 $\pm$ 7.06	0
Bituberous width (mm)	132.24 $\pm$ 9.15	136.46 $\pm$ 8.29	127.30 $\pm$ 7.49	0
Iliopectineal eminence width (mm)	102.38 $\pm$ 7.75	106.20 $\pm$ 7.02	97.90 $\pm$ 5.96	0
Pelvic inlet transverse (mm)	131.58 $\pm$ 8.88	134.85 $\pm$ 8.59	127.75 $\pm$ 7.61	0
Sacral width (mm)	117.08 $\pm$ 6.08	116.75 $\pm$ 5.67	117.47 $\pm$ 6.54	0
Subpubic angle (deg)	78.12 $\pm$ 10.12	85.80 $\pm$ 6.14	69.11 $\pm$ 5.26	0
Global scale factor (-)	0.96 $\pm$ 0.06	0.92 $\pm$ 0.05	1.00 $\pm$ 0.05	0
Total pelvic volume (cm ³)	823.58 $\pm$ 149.32	736.79 $\pm$ 105.99	925.28 $\pm$ 126.99	0
Total pelvic surface (cm ²)	1568.63 $\pm$ 169.26	1479.09 $\pm$ 133.64	1673.57 $\pm$ 144.80	0

2.2 Finite-element model

The archived simulation metadata specify DOLFINx 0.10, 166,808 cells and 48,291 mesh nodes. Small-strain linear elasticity was solved with PETSc on a shared reference domain. Subject anatomy was represented by $\mathbf{x} = \mathbf{X} + \phi_i(\mathbf{X})$; the displacement map was interpolated using radial basis functions (RBFs). Spatial gradients and integration measures were pulled back to the reference domain (Bonet and Wood, 1997). This representation approximates individual anatomy on a shared discretization; it does not guarantee exact cortical boundaries or invertibility without Jacobian checks.

The implemented bone-modulus law was

$$E/\text{MPa} = \begin{cases} 2398, & \rho_{\text{ash}} \leq 0.486, \\ 10200\rho_{\text{ash}}^2, & \rho_{\text{ash}} > 0.486, \end{cases} \quad \rho_{\text{ash}} = (\rho + 0.09)/1.14, \quad (1)$$

where numerical density values are in g cm⁻³ and ρ denotes the mapped hydroxyapatite-density input. This implemented approximation has a constant low-density branch, which suppresses modulus variation below its threshold; it must not be interpreted as an unrestricted test of bone quality (Keller, 1994). Bone Poisson ratio was 0.30. Both SIJ cartilage and pubic symphysis used $E = 1.0$ MPa and $\nu = 0.45$, as recorded in all three simulation archives. These nominal

choices were not calibrated to individual subjects.

Ligaments were assembled as linear spring stiffness matrices with a geometric stiffness contribution from prescribed pretension and a pretension force vector. The implementation does not update a tension-only active set during the static solve. Bilateral anterior SIJ, posterior SIJ, interosseous, sacrospinous and sacrotuberous bundles and the pubic bundle were included. Per-bundle stiffnesses were respectively 1400, 5600, 200, 3000, 3000 and 1000 N/mm; total pretension was 20 N per bundle. Each bundle’s parameters were divided among its segments. Bone–cartilage interfaces were conforming; frictional contact and muscle activation were not modeled.

2.3 Loads and constraints

The superior S1 facet was constrained using a surface penalty $\gamma = 10^6$ N/mm³ (Appendix A). Table 2 gives force vectors in the simulation’s global coordinates; these must be distinguished from the subsequently constructed pelvis-aligned reporting frame. SP2leg applies +400 N vertically to each acetabulum, and SP1leg applies +800 N to the right acetabulum. LAB1 applies opposing ML forces of ± 400 N to the internal ring surfaces; LAB2 applies the same signed pair to the ischial tuberosities; LAB3 applies opposing AP forces of ± 400 N at the pubis and sacrococcygeal junction. Inspection of the archived template places both left application patches at greater global x than their right counterparts; thus both ML pairs point outward in that template, contrary to the earlier description of LAB1 as compression. We name the cases by application site because a force direction does not establish the direction of joint motion.

Surface tractions were normalized by mapped surface area, giving identical prescribed force resultants across subjects. LAB force pairs have zero net force, whereas standing cases have an 800 N net vertical force balanced by the sacral constraint. Equal sums of force magnitudes do not imply equal moments, work, or physiological demand. The analyses compare responses to these specified load systems, not force-normalized compliance across equivalent activities. All load cases include the prescribed ligament pretension.

2.4 Reference configuration and SIJ kinematics

The corrected analysis compares each unloaded mapped surface $\mathbf{x}_i^0 = \mathbf{X} + \phi_i$ with its loaded surface $\mathbf{x}_{i\ell}^1 = \mathbf{x}_i^0 + \mathbf{u}_{i\ell}$. Stored FE displacement arrays were reprocessed without rerunning equilibrium solves. Node ordering was checked against all five archived reference displacement

Table 2: Standardized loads in global simulation coordinates. LAB1–LAB3 are independent mechanical proxies.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Internal ring force pair	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial force pair	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP force pair	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

fields. Geometry and displacement were evaluated with the same RBF parameters used by the original workflow. Checkpoints preserve subject row indices consistently across the three variants. Pairing follows the common indexed shape/density inputs and the archived processing order; the auxiliary archives do not contain an independent subject-ID manifest.

Relative ilium-to-sacrum motion was estimated using trimmed Kabsch registration of opposing surface regions within 5 mm, with a minimum of 80 nodes per region. Before registration, the inverse least-squares affine sacral map was applied to all loaded points. This removes sacral affine strain as well as rigid drift; the resulting quantities are affine-corrected relative motion, rather than unmodified physical joint displacement. Sensitivity to rigid-only drift removal remains necessary for external validation.

Motion was expressed in a pelvis-aligned ML/AP/CC frame, not a local articular normal/tangent frame. Fixed-axis Cardan xyz angles satisfy $R = R_z R_y R_x$. The bilateral mean of the Euclidean norms of these angle triplets is denoted $|\boldsymbol{\theta}|$; this is a small-angle summary, not the exact invariant rotation angle. Translation $|\mathbf{d}|$ is the bilateral mean of the norms of relative displacements evaluated at the sacral ROI centroid. Component summaries are bilateral means of absolute values. In particular, $|\theta_x|$ does not distinguish nutation from counternutation, and $|d_{ML}|$ is not a measurement of joint gapping. Translation-magnitude asymmetry is $||\mathbf{d}_L|| - ||\mathbf{d}_R||$.

2.5 Matched variants and variance ratios

The full variant uses subject-specific geometry and bone density. The shape-only variant uses subject geometry and the spatially varying pointwise population-median density field; it is not homogeneous bone. The material-only variant uses subject density on the template geometry. Cartilage and ligament parameters remain nominal in all variants.

For each load and endpoint we calculated $100V_{shape}/V_{full}$ and $100V_{material}/V_{full}$, with 2000 paired subject bootstrap samples (Efron and Tibshirani, 1994). These non-additive ratios can

exceed 100% and neither estimate causal variance fractions nor separate geometry–material interactions. Shape/full agreement was also assessed by RMSE, MAE and identity-line $R^2 = 1 - \sum_i (Y_{shape,i} - Y_{full,i})^2 / \sum_i (Y_{full,i} - \bar{Y}_{full})^2$.

2.6 Statistical analysis

Distributions are summarized by medians and interquartile ranges. Paired component contrasts use $\Delta_{i\ell,k} = |d_k|_{i\ell} - |d_k|_{i,SP2leg}$, with 3000 bootstrap samples for median confidence intervals and two-sided exact binomial sign tests of the median contrast (zero differences excluded). Wilcoxon signed-rank statistics are retained in the machine-readable output as a sensitivity diagnostic, but are not used for the primary inference because asymmetric differences can yield a signed-rank result that disagrees with the direction of the median. Benjamini–Hochberg correction was applied to the 12 component comparisons (Benjamini and Hochberg, 1995). H1 additionally uses paired tests for rotation magnitude, translation magnitude and translation asymmetry (a separate three-test family).

For each LAB case, nested OLS models used HC3 covariance (MacKinnon and White, 1985):

$$M0 : Y = \beta_0 + \beta_s \text{Female} + \beta_a \text{Age}_z + \epsilon, \quad (2)$$

$$M1 : Y = M0 + \beta_v \log(V)_z, \quad (3)$$

$$M2 : Y = M1 + \beta_{AP} \text{AP}_z + \beta_b \text{Biischadic}_z + \beta_p \text{Subpubic}_z. \quad (4)$$

Here model addition denotes addition to the linear predictor. Female is coded 1 and male 0; continuous predictors are standardized using cohort sample SDs. M2 contains no sex–volume interaction. FDR correction covers its six sex coefficients (two outcomes by three loads). M0/M1 and coefficient attenuation are descriptive. VIFs quantify collinearity without treating a threshold as proof of identifiability. The triad describes inlet depth, interspinous midpelvic width and the anterior pubic arch; it was selected for this retrospective analysis. Pooled and sex-stratified Spearman correlations are exploratory, not evidence of mediation.

Pooled OLS models include load, sex, age, log volume, the morphometric triad, load–volume interactions and a sex–volume interaction, with covariance clustered by the 278 subjects. A main volume coefficient therefore refers to male SP2leg, not an overall pooled slope. FDR is applied within each outcome’s pooled coefficients. Robust Wald statistics are not converted to partial variance-explained measures.

Exploratory load-specific models regress $\log Y$ on $\log V$, sex, their interaction, age and the triad. Male and female slopes are β_v and $\beta_v + \beta_{v:s}$; HC3 covariance supplies 95% Wald intervals. FDR covers 20 sex-specific slope tests, with a separate family of ten interaction tests. Failure to reject an interaction is not evidence of equal slopes. The alternative $\log(\text{scale})$ model is an algebraic reparameterization because $\log V = 3 \log(\text{scale}) + \log V_{\text{template}}$; it is not an independent robustness test. Sample confidence intervals do not include uncertainty in loads, constitutive choices, segmentation or boundary conditions.

3 Results

3.1 Cohort and corrected extraction

The corrected dataset contains 278 subjects and 1390 subject-load observations in the full variant, with corresponding records in both auxiliary variants. Cohort demographics and dimensions are unchanged (Table 1). All numerical results below use the unloaded subject reference. The unloaded-to-unloaded checks were below 10^{-8} in degrees and mm for every subject.

3.2 Standing load contrasts (H1)

Under SP2leg and SP1leg, median rotation norms were 0.956° and 2.040° , and median translations were 1.058 and 1.627 mm, respectively. Median bilateral translation asymmetry was 0.065 mm in SP2leg and 1.602 mm in SP1leg (Fig. 2). The paired SP1leg-minus-SP2leg median difference in rotation norm was $+1.048$ (95% CI $+1.004$ to $+1.102$) degrees, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation magnitude was $+0.582$ (95% CI $+0.559$ to $+0.603$) mm, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation asymmetry was $+1.509$ (95% CI $+1.445$ to $+1.568$) mm, $q < 0.001$. These comparisons test motion and asymmetry; absolute ML rotation does not establish self-bracing nutation.

Table 3: Corrected SIJ motion summaries across 278 subjects. Values are median [interquartile range]; component columns are absolute magnitudes.

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	0.956 [0.813, 1.109]	1.058 [0.871, 1.207]	0.366 [0.241, 0.482]	0.083 [0.054, 0.123]	0.952 [0.818, 1.112]
SP1leg	278	2.040 [1.765, 2.289]	1.627 [1.430, 1.869]	0.440 [0.357, 0.520]	0.235 [0.164, 0.345]	1.428 [1.213, 1.644]
LAB1	278	0.221 [0.188, 0.268]	0.154 [0.121, 0.190]	0.060 [0.044, 0.079]	0.077 [0.063, 0.092]	0.105 [0.073, 0.144]
LAB2	278	0.724 [0.649, 0.812]	0.255 [0.204, 0.328]	0.176 [0.099, 0.262]	0.134 [0.115, 0.154]	0.075 [0.049, 0.118]
LAB3	278	0.844 [0.672, 1.057]	0.574 [0.448, 0.707]	0.372 [0.260, 0.515]	0.092 [0.069, 0.115]	0.387 [0.326, 0.477]

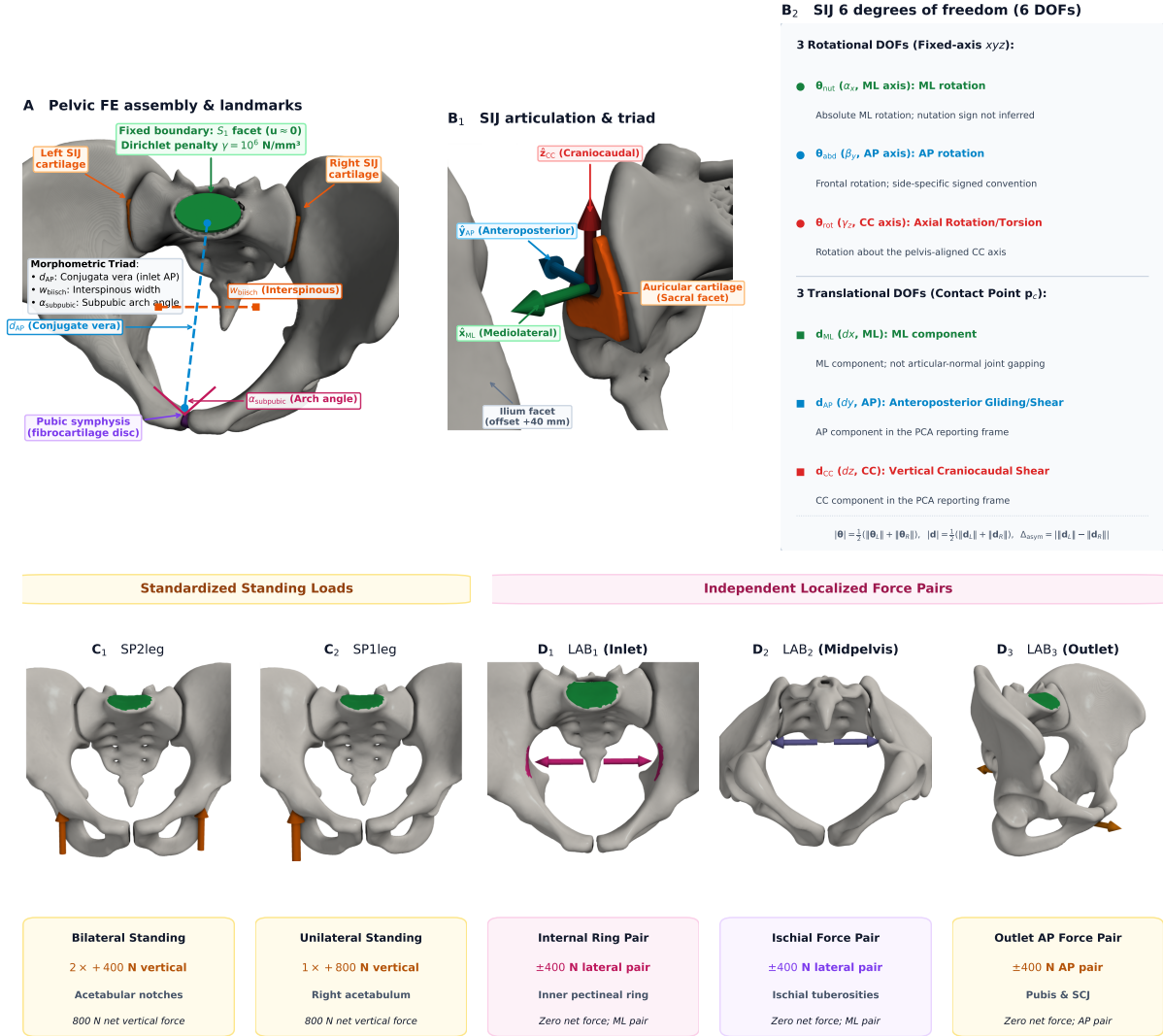


Figure 1: Model anatomy and load definitions. (A) Pelvic assembly with schematic morphometric overlays; their drawn locations illustrate dimensions and are not quantitative landmark measurements. Green denotes the S1 constraint, orange the SIJ cartilage and purple the symphysis. (B) Exploded articulation and pelvis-aligned coordinate definitions; arrows illustrate axes, not measured trajectories. Rotations describe ilium relative to sacrum. (C) Bilateral and unilateral acetabular loads. (D) Independent ring, ischial and AP force pairs. Arrows represent applied global force directions; they do not demonstrate joint opening. The penalty approximates a fixed boundary.

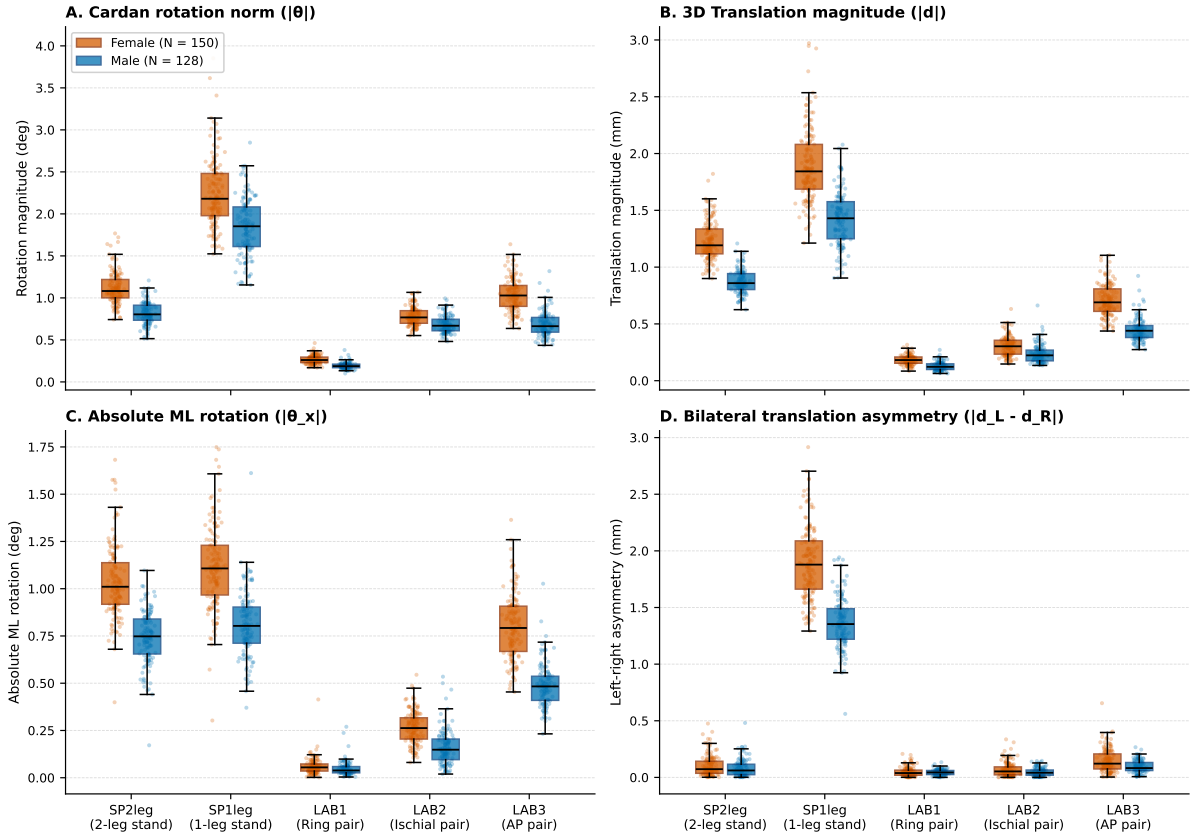


Figure 2: **Corrected motion distributions.** Rotation-triplet norm, relative translation norm, absolute ML rotation and bilateral translation-magnitude asymmetry. Points are subjects; box-plots show medians and IQRs, with whiskers at 1.5 IQR. Female and male groups use vermilion and blue. Jitter is deterministic. No signed nutation direction is inferred.

3.3 Component contrasts (H2)

LAB1: the ML absolute-component contrast was -0.007 (95% CI -0.018 to $+0.002$) mm, $q = 0.254$; the AP absolute-component contrast was -0.303 (95% CI -0.322 to -0.270) mm, $q < 0.001$; the CC absolute-component contrast was -0.854 (95% CI -0.886 to -0.817) mm, $q < 0.001$.

LAB2: the ML absolute-component contrast was $+0.052$ (95% CI $+0.047$ to $+0.061$) mm, $q < 0.001$; the AP absolute-component contrast was -0.181 (95% CI -0.207 to -0.157) mm, $q < 0.001$; the CC absolute-component contrast was -0.877 (95% CI -0.900 to -0.841) mm, $q < 0.001$.

LAB3: the ML absolute-component contrast was $+0.011$ (95% CI $+0.003$ to $+0.017$) mm, $q = 0.007$; the AP absolute-component contrast was $+0.032$ (95% CI $+0.018$ to $+0.043$) mm, $q < 0.001$; the CC absolute-component contrast was -0.556 (95% CI -0.586 to -0.537) mm, $q < 0.001$.

All contrasts are paired differences in absolute magnitudes relative to SP2leg (Fig. 3; Supplementary Table S1). The state-space arrows connect the baseline to componentwise median contrasts; they are not physical displacement vectors or trajectories.

3.4 Sex and morphometric associations (H3)

LAB1 female-minus-male translation coefficients were $+0.055$, $+0.040$ and $+0.013$ mm in M0, M1 and M2. The M2 interval was -0.005 to $+0.031$ mm ($p = 0.166$, $q = 0.566$). LAB2 female-minus-male translation coefficients were $+0.071$, $+0.041$ and -0.031 mm in M0, M1 and M2. The M2 interval was -0.078 to $+0.016$ mm ($p = 0.192$, $q = 0.566$). LAB3 female-minus-male translation coefficients were $+0.258$, $+0.212$ and $+0.002$ mm in M0, M1 and M2. The M2 interval was -0.060 to $+0.064$ mm ($p = 0.950$, $q = 0.950$). The maximum non-intercept VIF in M2 was 5.55. For LAB1, the pooled Spearman correlation between subpubic angle and translation magnitude was $r_s = 0.444$ ($p < 0.001$). The male-only correlation was $r_s = -0.063$ ($p = 0.480$). The female-only correlation was $r_s = -0.145$ ($p = 0.077$). These exploratory correlations are uncorrected for multiplicity and do not estimate a geometric mechanism. Figure 4 and Supplementary Table S3 report M2 estimates for both outcomes.

3.5 Restricted geometry/material comparison (H4)

Across five loads and five endpoints, shape-only/full variance ratios ranged from 96.43% to 120.61% (median 100.68%). Material-only/full ratios ranged from 0.014% to 5.789% (median

0.244%). For rotation, minimum shape/full identity-line R^2 was 0.8777, and maximum RMSE was 0.0318 degrees. For translation, minimum shape/full identity-line R^2 was 0.9457, and maximum RMSE was 0.0280 mm. These compare model variants and do not decompose an additive population variance budget (Fig. 5; Supplementary Table S2).

3.6 Exploratory volume associations

20 of 20 sex-specific slopes survived FDR correction. The exponents ranged from -1.062 to -0.366; the largest adjusted slope q was 0.002. Thus negative conditional volume associations extended across standing and all three localized force-pair configurations, rather than being restricted to unilateral standing. Full estimates and intervals are in Table 4. Sex-by-volume interaction q values ranged from 0.343 to 0.699; nonsignificance does not establish equivalence. Figure 6 and Table 4 use the same adjusted models. Pooled cluster-robust coefficients are reported in Supplementary Table S5.

Table 4: Adjusted log–log exponents by load and sex. Slope q values use the 20-test family; interaction q values use a separate ten-test family.

Load	Outcome	Male exponent [95% CI]	Male q	Female exponent [95% CI]	Female q	Interaction q
SP2leg	Rotation (deg)	-0.679 [-0.890, -0.469]	< 0.001	-0.605 [-0.764, -0.446]	< 0.001	0.699
SP2leg	Translation (mm)	-0.366 [-0.531, -0.200]	< 0.001	-0.402 [-0.522, -0.282]	< 0.001	0.699
SP1leg	Rotation (deg)	-0.622 [-0.895, -0.349]	< 0.001	-0.550 [-0.760, -0.339]	< 0.001	0.699
SP1leg	Translation (mm)	-0.460 [-0.711, -0.208]	< 0.001	-0.397 [-0.591, -0.203]	< 0.001	0.699
LAB1	Rotation (deg)	-0.916 [-1.144, -0.687]	< 0.001	-1.062 [-1.233, -0.891]	< 0.001	0.588
LAB1	Translation (mm)	-0.568 [-0.925, -0.210]	0.002	-1.003 [-1.270, -0.737]	< 0.001	0.343
LAB2	Rotation (deg)	-0.541 [-0.714, -0.367]	< 0.001	-0.674 [-0.804, -0.544]	< 0.001	0.494
LAB2	Translation (mm)	-0.818 [-1.179, -0.456]	< 0.001	-0.947 [-1.287, -0.606]	< 0.001	0.699
LAB3	Rotation (deg)	-0.850 [-1.048, -0.651]	< 0.001	-1.045 [-1.240, -0.851]	< 0.001	0.494
LAB3	Translation (mm)	-0.700 [-0.922, -0.478]	< 0.001	-0.904 [-1.159, -0.649]	< 0.001	0.494

3.7 Extraction sensitivity and hypothesis assessment

In six subjects selected at approximately evenly spaced archive indices (30 subject–load comparisons), replacing affine sacral correction with rigid-only correction changed bilateral rotation norms by at most 0.0229° and translation magnitudes by at most 0.0055 mm. This subset diagnostic does not bound errors in the remaining cohort (Supplementary Table S6).

- **H1:** Supported for the specified static endpoints. This does not verify a nutation or force-closure mechanism.

- **H2:** The full proposed LAB1/LAB2 component pattern is not supported after multiplicity correction. LAB3 is a separate contrast; signed opening remains untested.
- **H3:** No positive fully adjusted female translation association survived FDR correction. The nested coefficient changes are descriptive and do not establish causal mediation.
- **H4:** Shape/full variance retention and paired errors support only the stated comparison with a reference bone-density field. They do not rank geometry against unmodeled soft-tissue variability.

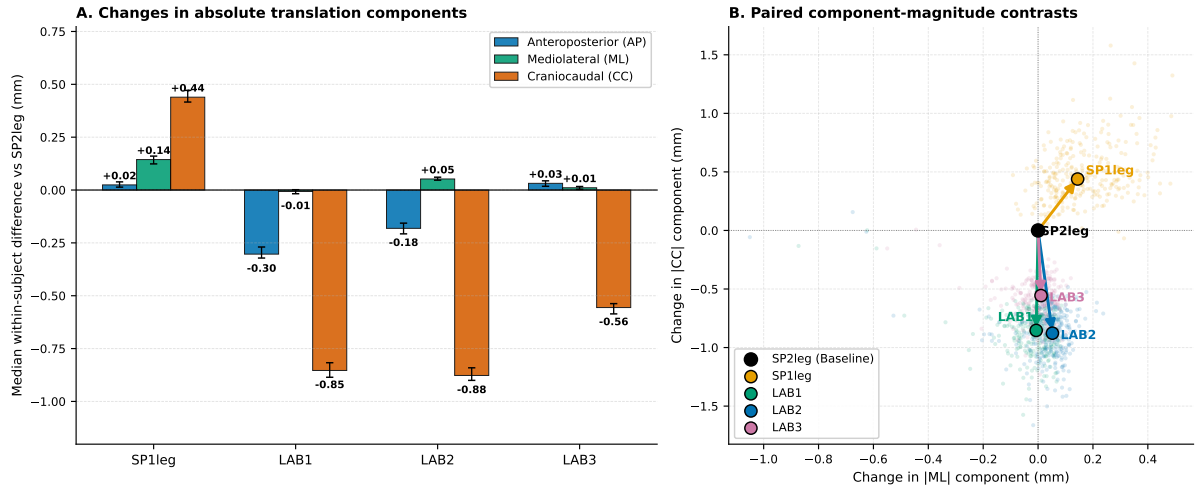


Figure 3: **Paired absolute-component contrasts.** (A) Median within-subject differences from SP2leg, with bootstrap 95% CIs. (B) Individual ML/CC contrast pairs and arrows to their componentwise medians. Positive values mean larger component magnitude, not outward joint motion. Corrected q values are reported in Supplementary Table S1; intervals are pointwise, not simultaneous.

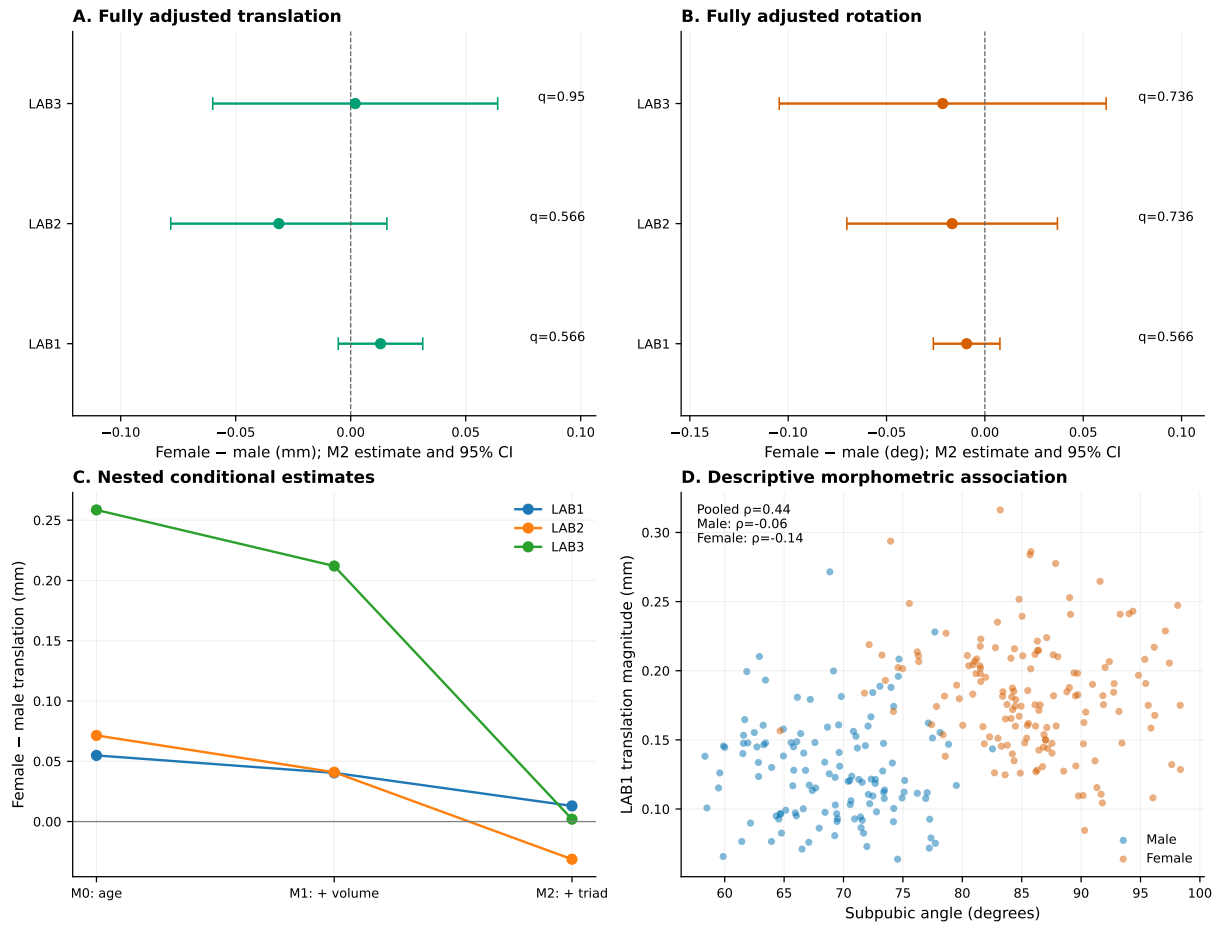


Figure 4: **Conditional sex estimates and morphology.** (A,B) M2 female-minus-male coefficients with HC3 95% CIs, shown on separate translation and rotation axes. Labels give BH-adjusted q values across six tests. (C) Nested translation coefficients; connecting lines show changes between model specifications, not causal pathways. (D) LAB1 translation versus subpubic angle, with pooled and sex-stratified Spearman correlations. All panels use corrected endpoints.

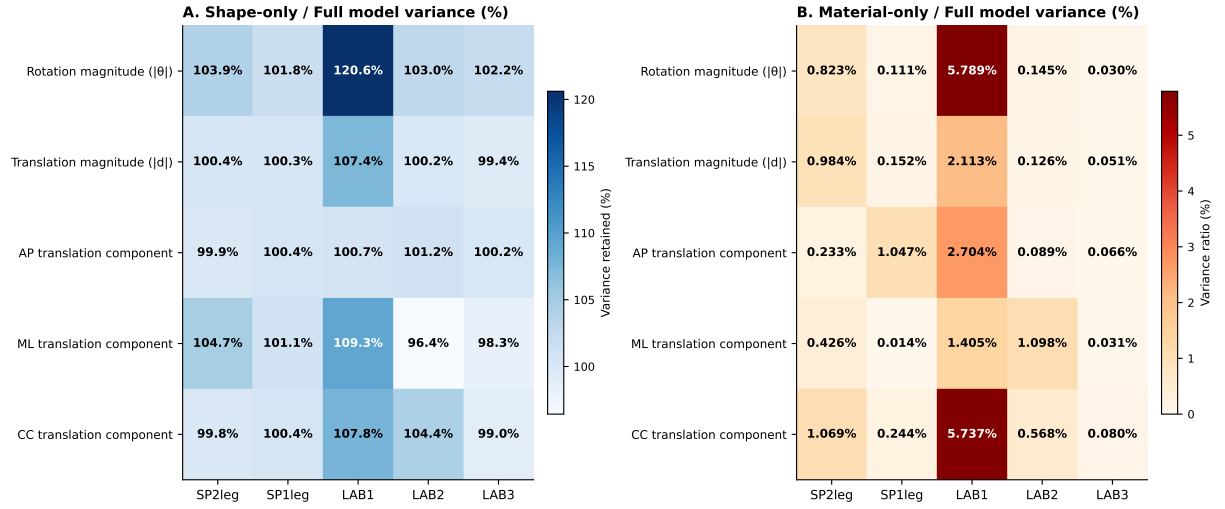


Figure 5: **Variance ratios across matched variants.** (A) Shape-only/full and (B) material-only/full sample-variance ratios, in percent. Five endpoints are shown. Color scales differ between panels and follow their actual data ranges. Values above 100% are allowed. Bootstrap intervals and paired agreement diagnostics are provided in Supplementary Table S2; neither panel represents variance explained by an independent causal factor.

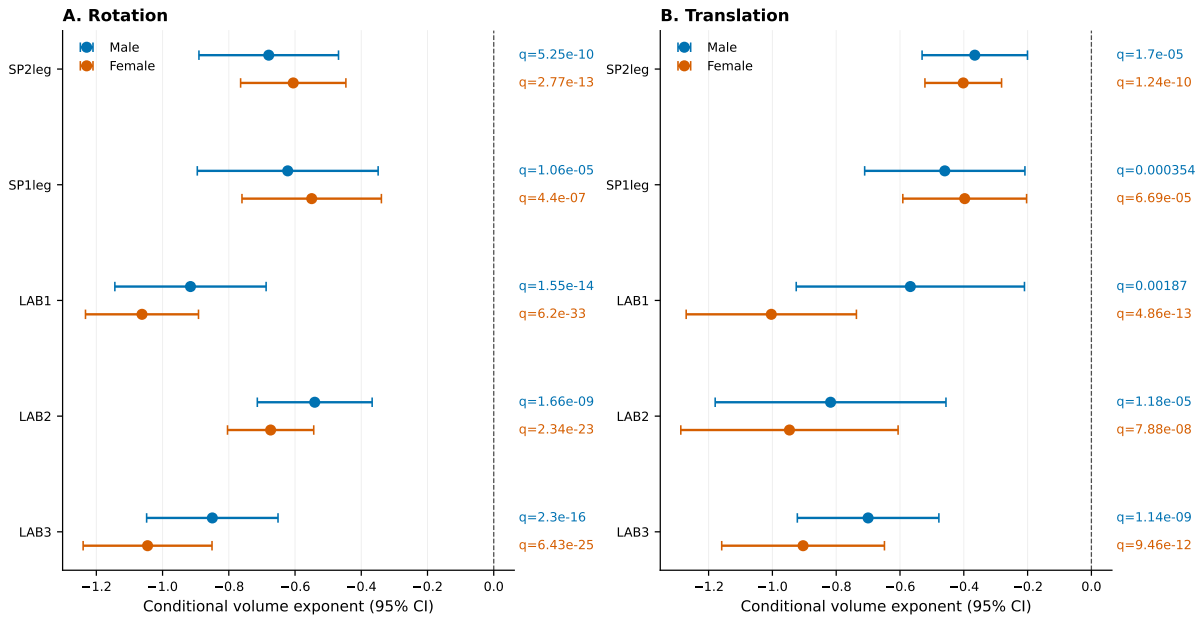


Figure 6: **Conditional log-volume slopes.** Male and female exponents with HC3 95% CIs from the same adjusted log-log models as Table 4. The vertical zero line indicates no conditional association. Confidence intervals are pointwise; labels give slope q values from the 20-test family. Raw scatterplots and adjusted conditional geometric-mean curves are supplied as Supplementary Figure S1.

4 Discussion

Across 278 pelvic models, unilateral standing produced larger relative joint motion and bilateral asymmetry than bilateral standing. The localized force-pair responses differed by application site; they did not support one common transverse-expansion pattern. Female translation coefficients were positive in the age-adjusted models, but no residual female translation association survived full adjustment and FDR correction. Negative conditional volume slopes occurred across all five loads. These findings support load-specific structural comparisons, rather than a general female compliance surplus independent of measured anatomy.

Reference configuration and credibility of the endpoints

Elastic motion was measured against each subject’s unloaded mapped anatomy. This distinction matters in population FE models: comparing a shared template directly with a loaded individual would include anatomical variation in the endpoint, potentially inflating associations with geometry. The unloaded-to-unloaded checks and saved-field reconstruction checks provide numerical verification of the extraction, not experimental validation of the equilibrium model.

The affine sacral correction remains a modeling convention rather than only a rigid change of coordinates. Removing sacral strain can change fitted relative motion. The sensitivity analysis found small differences from rigid-only correction in six subjects, but it does not establish cohort-wide external validity. Reported confidence intervals reflect subject sampling and do not cover this extraction choice or FE model discrepancy.

Load dependence and anatomical interpretation

Unilateral acetabular loading supplies a distinct asymmetric load path, so a paired comparison with bilateral loading directly addresses a mechanical question. However, this static configuration omits the time-varying muscle forces and support conditions of gait. Relative rotation about the ML axis is compatible with sagittal pelvic motion, but its absolute value cannot distinguish nutation from counternutation. Neither ligament tension changes nor frictional self-bracing were measured here. The results therefore do not independently verify force closure, despite its established anatomical relevance (Snijders et al., 1993; Vleeming et al., 2012).

The LAB cases apply force pairs at different sites and should be interpreted individually. A reduction in $|d_{CC}|$ accompanied by an increase in $|d_{ML}|$ describes redistribution among the

chosen axes. It does not establish increased birth-canal dimensions, preservation of joint congruency or a transition from compression to opening. Such claims would require signed motion in anatomically anchored frames, distances between opposing surfaces and direct inlet, midplane or outlet dimension changes. In particular, ML displacement is not generally normal to the auricular surface. Static force pairs omit fetal contact, pelvic floor mechanics, changing posture and pregnancy-related tissue adaptation (Chen and Grimm, 2021).

Sex and morphology are associated, not experimentally separated

The nested regressions show that the initial positive sex coefficients attenuate substantially with measured dimensions; LAB2 changes sign and all three fully adjusted translation intervals include zero. This removes the basis for claiming a persistent female translational surplus independent of the included anatomy. Attenuation is not a percentage of biological mediation: sex, dimensions and unmeasured anatomy are related, covariates are measured with error, and conditional comparisons may have limited support where male and female morphologies overlap poorly. The positive pooled subpubic-angle correlation contrasts with small negative within-sex correlations (Fig. 4), consistent with group separation contributing to the pooled association. No hoop-stiffness mechanism was directly measured. Residual sex coefficients cannot distinguish articular orientation, omitted dimensions, mapping error or other structural differences.

What the material comparison establishes

Shape-only/full variance ratios were near 100% for many endpoints but reached 120.61%, and rotation identity-line R^2 fell to 0.878. Thus a reference bone-density field often reproduces the broad between-subject dispersion, but its accuracy is endpoint-dependent and is not uniformly near-perfect. It would not show that mineralization is biologically irrelevant. The implemented modulus law is constant below its density threshold, soft-tissue properties are fixed, and material-only responses are evaluated at one reference geometry. Geometry-material interactions are therefore incompletely sampled. Ligament pretension has been shown to alter SIJ predictions in other FE models (Heyland et al., 2025); its contribution is not bounded by the small bone-density-only variance ratio found here.

Variance ratios above 100% are mathematically permissible because the variants are separate deterministic systems. They are not percentages of an additive variance budget. Similarly, a high identity-line R^2 is an internal agreement measure, not validation against physical mea-

surements. The repaired extraction removes one source of artificial geometry dominance, but external validation and constitutive sensitivity are still required.

Size scaling under prescribed force

Negative volume slopes need not imply a distinct evolutionary adaptation. For geometrically similar linear-elastic solids with fixed modulus and a fixed resultant force, dimensional scaling gives $u \propto L^{-1}$ and $\theta \sim u/L \propto L^{-2}$, corresponding to volume exponents $-1/3$ and $-2/3$. This is a dimensional benchmark, not a prediction for the present heterogeneous, jointed pelvis with fixed spring parameters. The fitted models additionally condition on linear dimensions, so their volume coefficients describe residual size associations rather than uniform geometric scaling. Forces proportional to body weight or area would give different benchmarks. A nonsignificant sex-by-size interaction does not establish equal slopes; equivalence requires a prespecified tolerance and a suitably powered test.

Originality, usefulness and limits of generalization

The main potential contribution beyond the earlier morphology study (Henyš and Hammer, 2025) is a paired map of load-dependent joint responses, coupled to a reproducible comparison of geometry and bone-density variants. Neither a large cohort nor the general importance of geometry is sufficient as a novelty claim. The degree of imaging overlap with previous work must be documented; overlapping anatomy does not constitute independent validation.

The immediate use is methodological: selecting load cases for subsequent experiments, describing distributions of model response, and assessing whether individual bone-density mapping materially changes the selected outputs. No pain data, fixation models, delivery outcomes, parity or pregnancy-specific material measurements were analyzed. The results cannot recommend surgical fixation directions, select a delivery mode, diagnose instability or define normal clinical thresholds.

Absolute magnitudes should be compared with experiments only when loading, reference configuration and endpoint definitions match. Small SIJ movements are challenging to measure, and published estimates depend strongly on methods (Goode et al., 2008). The 16–91-year cohort is not a cohort of laboring women. Its complete records do not guarantee population representativeness. The fixed sacrum, nominal cartilage, linearized ligaments, absence of contact and muscles, RBF approximation, ROI definition and affine correction all constrain interpretation.

Mesh convergence, mapped Jacobian quality, stiffness/pretension uncertainty and independent
cadaveric or in-vivo validation should accompany any claim of quantitative physiological predic-
tion.

5 Conclusion

In 278 pelvic finite-element models, unilateral acetabular loading increased joint motion and
bilateral translation asymmetry relative to bilateral loading. Localized force pairs produced dis-
tinct component responses, rather than a shared pattern of transverse opening. No fully adjusted
female translation association survived multiplicity correction, while conditional volume slopes
were negative across all five loads. Shape-only models broadly retained full-model variability,
with endpoint-dependent individual errors. These findings support reproducible comparisons of
specified passive mechanical configurations. Claims about obstetric adaptation, force closure,
pain or surgical benefit require measurements not provided by these simulations.

A Mathematical formulation and verification

A.1 Reference mapping and equilibrium

Let Ω_0 be the FE template and $\chi_i(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X})$ the unloaded subject geometry. The
anatomical mapping and the elastic displacement are different fields. With $\mathbf{F}_i = \mathbf{I} + \nabla_X \phi_i$ and
 $J_i = \det \mathbf{F}_i$, the physical small-strain tensor is

$$\boldsymbol{\varepsilon}_i(\mathbf{u}) = \text{sym}(\nabla_X \mathbf{u} \mathbf{F}_i^{-1}). \quad (5)$$

The pullback requires an orientation-preserving, nonsingular map. RBF smoothness alone does
not prove this condition. For physical coordinates in mm, the discrete weak problem is

$$\int_{\Omega_i} \mathbf{C}_i \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) dV + \int_{\Gamma_{S1,i}} \gamma \mathbf{u} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{K}_{lig,i} \mathbf{u}_h = \int_{\Gamma_{load,i}} \mathbf{t} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{f}_{pre,i}. \quad (6)$$

The surface penalty has units N/mm³, because its associated traction is $\gamma \mathbf{u}$ in N/mm². It
imposes an approximate fixed boundary, not exact $\mathbf{u} = 0$. The ligament stiffness and pretension
terms are required for consistency with the assembled solver. For one spring segment of length

369 L , direction $\mathbf{e}$, assigned stiffness k and pretension T , its directional stiffness is

$$370 \quad \mathbf{W} = k\mathbf{e}\mathbf{e}^T + (T/L)(\mathbf{I} - \mathbf{e}\mathbf{e}^T). \quad (7)$$

371 This matrix is assembled with opposite off-diagonal blocks at its two endpoints. It is a linearized
372 spring representation without tension-only switching.

373 The surface measure obeys $dS_i = J_{surf,i}dS_0$, where

$$374 \quad J_{surf,i} = J_i \|\mathbf{F}_i^{-T} \mathbf{n}_0\|, \quad \mathbf{t} = \frac{\mathbf{F}^{tot}}{\int_{\Gamma_{load,0}} J_{surf,i} dS_0}. \quad (8)$$

375 Thus the applied physical force integral equals $\mathbf{F}^{tot}$. The penalty and load integrals use the
376 corresponding mapped surface measures. The force vectors are listed in Table 2; their global
377 axes are not assumed identical to the PCA reporting axes below.

378 A.2 Unloaded and loaded surfaces

379 Let $\mathbf{X}_j$ be surface template nodes. The two point clouds used for kinematics are

$$380 \quad \mathbf{p}_{ij}^0 = \mathbf{X}_j + \phi_i(\mathbf{X}_j), \quad \mathbf{p}_{ilj}^1 = \mathbf{p}_{ij}^0 + \mathbf{u}_{il}(\mathbf{X}_j). \quad (9)$$

381 Both use identical topology and point correspondence. Comparing $\mathbf{X}_j$ with $\mathbf{p}_{ilj}^1$ would incor-
382 rectly include subject morphology in the measured motion. Geometry was reconstructed with
383 the original two-stage interpolation: template-to-FE mapping (10 neighbors, smoothing 50) fol-
384 lowed by FE-to-surface mapping (10 neighbors, zero smoothing). Saved elastic fields use the
385 latter interpolation. Cached sparse interpolation weights implement the same linear RBF eval-
386 uation; numerical comparisons against direct RBF evaluation check this equivalence.

387 A.3 Pelvis-aligned reporting frame

388 The origin $\mathbf{o}$ is the centroid of the unloaded surface. The initial ML axis is the dominant
389 eigenvector of the point-cloud covariance, with deterministic sign. The symmetry-plane routine
390 uses this PCA approximation; it does not optimize reflection symmetry. Project the centered
391 points onto its orthogonal plane and take the leading in-plane singular vector. Orthogonalization
392 gives $\hat{\mathbf{z}}$, followed by

$$393 \quad \hat{\mathbf{y}} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}, \quad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \quad \mathbf{P} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}]. \quad (10)$$

The ML and AP axes are simultaneously reversed if needed so that ML points from the left to right ilium centroids. This construction is orthonormal and right-handed but does not independently anchor the AP/CC signs to anatomical landmarks. Absolute components are therefore used for comparisons. No clinical nutation sign is inferred from the stored α_x sign.

A.4 Affine correction, regions and rigid fits

Fit the sacral point correspondence by ordinary least squares,

$$(\mathbf{A}_S, \mathbf{b}_S) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_{j \in S} \|\mathbf{p}_j^1 - \mathbf{A} \mathbf{p}_j^0 - \mathbf{b}\|^2, \quad \mathbf{q}_j = \mathbf{A}_S^+(\mathbf{p}_j^1 - \mathbf{b}_S), \quad (11)$$

where $+$ denotes the Moore–Penrose pseudoinverse. This removes fitted affine sacral deformation, including stretch and shear; it is not only a change of rigid observer frame.

For each side, select sacral and iliac points within 5 mm of the opposing point cloud. If fewer than 80 points qualify, use up to the nearest 80 available points. Distances are between discrete points rather than continuous articular surfaces. For each bone ROI, the rigid fit maps $\mathbf{p}^0$ to $\mathbf{q}$. With centered row matrices $\tilde{\mathbf{P}}_B, \tilde{\mathbf{Q}}_B$,

$$\tilde{\mathbf{P}}_B^T \tilde{\mathbf{Q}}_B = \mathbf{U} \Sigma \mathbf{V}^T, \quad \mathbf{R}_B = \mathbf{V} \text{diag}(1, 1, \det(\mathbf{V} \mathbf{U}^T)) \mathbf{U}^T, \quad \mathbf{t}_B = \bar{\mathbf{q}}_B - \mathbf{R}_B \bar{\mathbf{p}}_B^0. \quad (12)$$

The fit is repeated after excluding the largest 10% residuals, using two trimming iterations. Trimming reduces outlier influence but does not validate the ROI as a physical contact patch.

A.5 Relative rotation and translation

For sacral and iliac fits S, I ,

$$\mathbf{R}_{rel} = \mathbf{R}_S^T \mathbf{R}_I, \quad \mathbf{R}_{loc} = \mathbf{P}^T \mathbf{R}_{rel} \mathbf{P} = \mathbf{R}_z(\gamma_z) \mathbf{R}_y(\beta_y) \mathbf{R}_x(\alpha_x). \quad (13)$$

The fixed-axis xyz extraction, for zero-based matrix indices away from gimbal lock, is

$$\alpha_x = \text{atan2}(R_{21}, R_{22}), \quad \beta_y = \arcsin[-\text{clip}(R_{20}, -1, 1)], \quad \gamma_z = \text{atan2}(R_{10}, R_{00}). \quad (14)$$

At $\beta_y = \pm\pi/2$, choose $\gamma_z = 0$ and $\alpha_x = \text{atan2}[-\text{sign}(R_{20})R_{01}, R_{11}]$. In implementation, the equivalent stable expression $\beta_y = \text{atan2}(-R_{20}, \sqrt{R_{00}^2 + R_{10}^2})$ is used, with the singular branch

restricted to vanishing cosine. The singular branch is included for completeness; physiological small-angle outputs should remain far from it. Matrix reconstruction tests cover both branches.

For the unloaded sacral ROI centroid $\mathbf{p}_c$, the reported displacement is

$$\mathbf{d} = \mathbf{P}^T[(\mathbf{R}_I - \mathbf{R}_S)\mathbf{p}_c + \mathbf{t}_I - \mathbf{t}_S]. \quad (15)$$

It evaluates the difference of two fitted maps at a common reference point. It is neither the distance between opposing articular surfaces nor a displacement along a surface normal. The bilateral summaries are

$$|\boldsymbol{\theta}| = \frac{1}{2} \sum_{s=L,R} (\alpha_{x,s}^2 + \beta_{y,s}^2 + \gamma_{z,s}^2)^{1/2}, \quad |\mathbf{d}| = \frac{1}{2} \sum_{s=L,R} \|\mathbf{d}_s\|, \quad (16)$$

$$|d_k| = \frac{1}{2}(|d_{k,L}| + |d_{k,R}|), \quad A_d = \left| \|\mathbf{d}_L\| - \|\mathbf{d}_R\| \right|. \quad (17)$$

Angles are converted to degrees before reporting. The exact principal rotation angle would instead be $\arccos[(\text{tr } \mathbf{R}_{rel} - 1)/2]$ with numerical clipping. The Cardan norm approximates it at small angles and depends on the chosen axes and sequence.

A.6 Size and conditional scaling

The archived allometry workflow sums volumes of the selected left ilium, right ilium and sacrum surface bodies, using mapped surface geometry and converting mm^3 to cm^3 . These are whole-bone geometric volumes, not birth-canal volume or a direct integral of cortical tissue. The size factor is $s_i = (V_i/V_{template})^{1/3}$. Therefore a log-volume slope b becomes $3b$ under log scale, with identical fitted responses and tests.

For the sex-interaction log model, $b_M = \beta_v$ and $b_F = \beta_v + \beta_{v:s}$. The female slope variance is

$$\text{Var}(b_F) = \text{Var}(\beta_v) + \text{Var}(\beta_{v:s}) + 2 \text{Cov}(\beta_v, \beta_{v:s}). \quad (18)$$

Reported intervals use this full covariance. Exponentiating a fitted log response gives a conditional geometric mean, not an unbiased arithmetic mean. Figure 6 uses this same adjusted model, with the other continuous covariates set to their cohort means.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. Final author approval of this revised version is pending.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information provides statistical definitions, component contrasts, variance ratios, sex models, the scale reparameterization, and extraction sensitivity diagnostics.

Data Availability Statement

The manuscript working package contains corrected subject-level endpoints, statistical tables and figures. The original FE displacement archives are retained separately. A persistent public or reviewer-accessible repository link must be supplied before submission.

Code Availability Statement

The working package includes scripts for corrected endpoint extraction, statistical analysis and figure generation. A versioned code archive and environment specification must accompany the

data repository before submission.

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